TERM 4

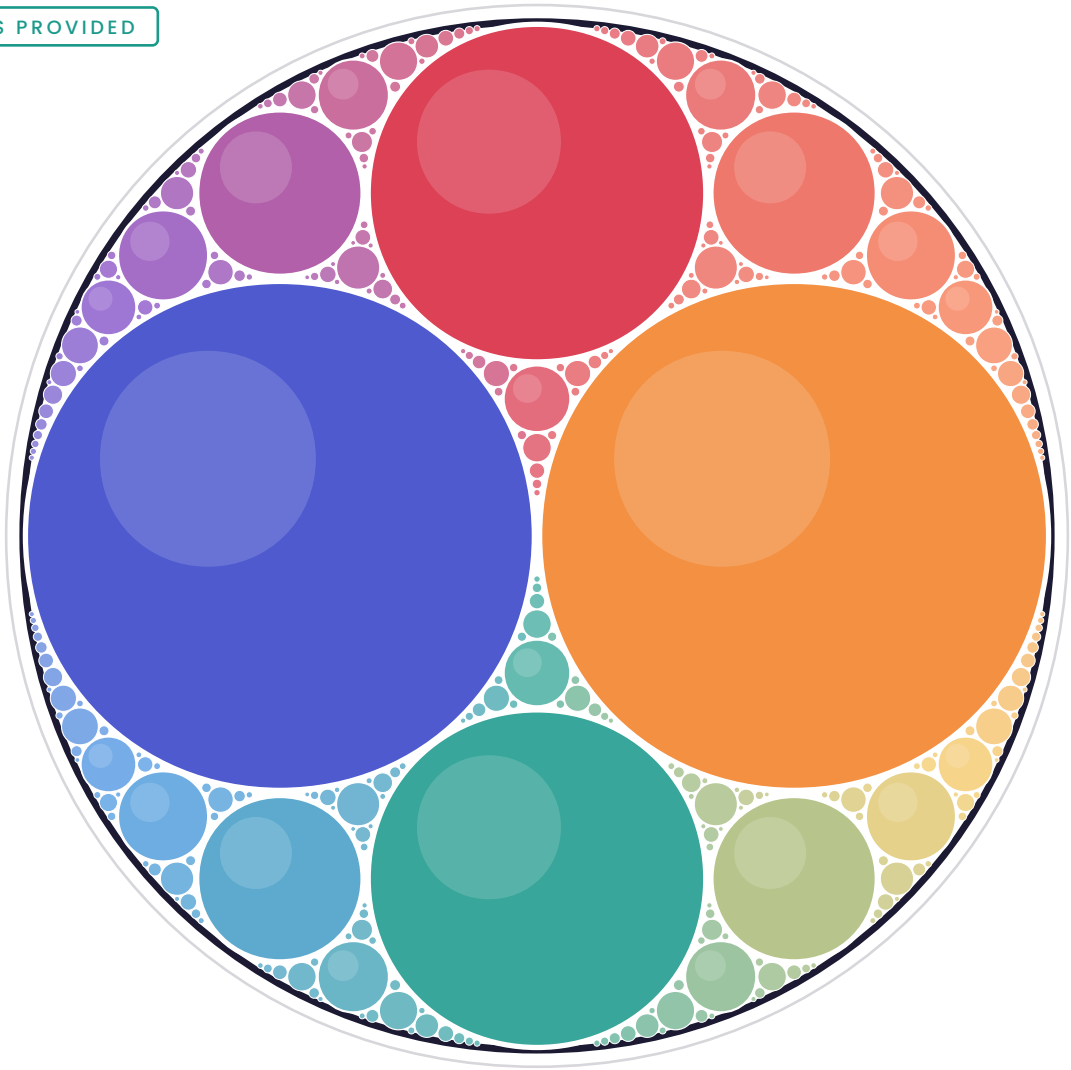
FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

YEAR 8

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

APOLLONIAN GASKET
every circle touches three others; every curvature is a whole number

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

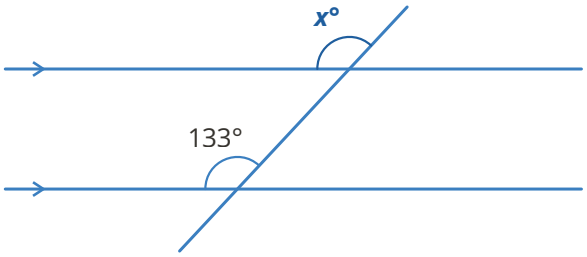
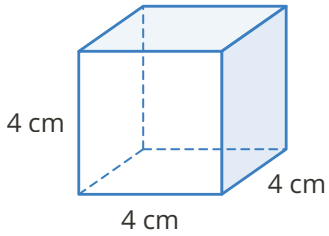
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Corresponding angles are equal COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism.  	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $-8 + 12 \div 4$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Share \$150 in the ratio 2 : 3. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 5 - x$. x | 0 | 1 | 2 | 3 y | 5 | 4 | |	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

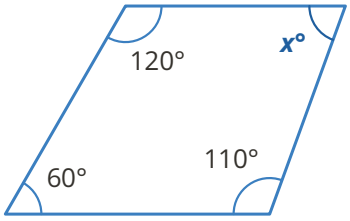
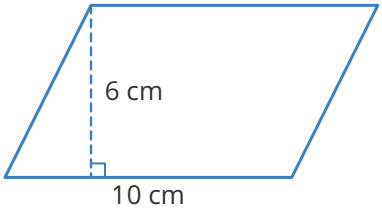
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 10% of $350	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Expand. $5(3x - 2)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the median of these scores. 12, 15, 11, 18, 14, 20, 16	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

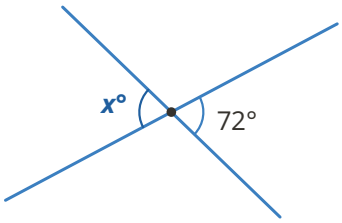
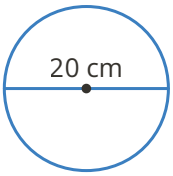
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 1 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 $\frac{3}{4}$ of the 32 nippers at the surf club passed the swim test. How many passed?	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Jo types 450 words in 9 minutes. Find Jo's typing rate in words per minute.	FORMULA Unit rate = amount ÷ number of units COPY SUBSTITUTE ANSWER
5 Find the probability that a goal is not scored. $P(\text{goal}) = \frac{3}{7}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

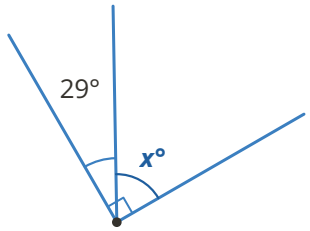
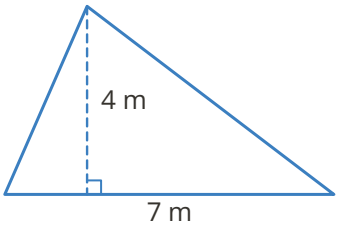
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the area of this triangle.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $\sqrt[3]{125}$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Solve. $4x - 7 = 13$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 Find the mean of these netball scores. Points: 12, 15, 9, 18, 6	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

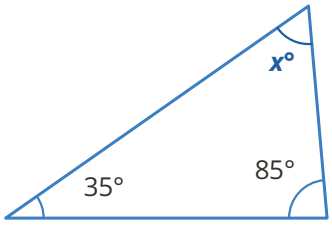
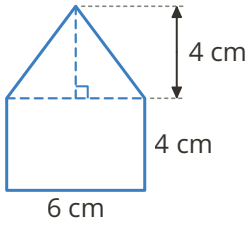
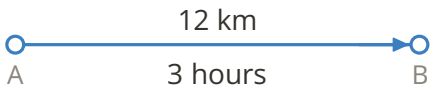
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Write as a decimal. $\frac{7}{20}$	FORMULA Fraction to decimal: numerator ÷ denominator COPY SUBSTITUTE ANSWER
4 A kayaker paddles from A to B. Find the average speed. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Heads came up 27 times in 90 tosses of a coin. Find the experimental probability of heads.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

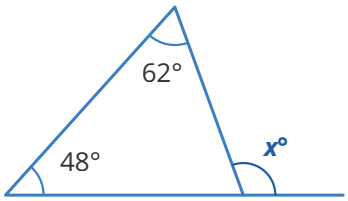
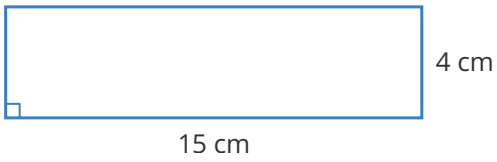
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 2 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this rectangle.  	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 12 of the 16 nippers at the surf club passed the swim test. What percentage passed?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Find the value of this expression when t = 3. $6 + 2t^2$	FORMULA $6 + 2t^2$ COPY SUBSTITUTE ANSWER
5 Find the range of these maximum temperatures. °C: 19, 25, 22, 31, 27	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

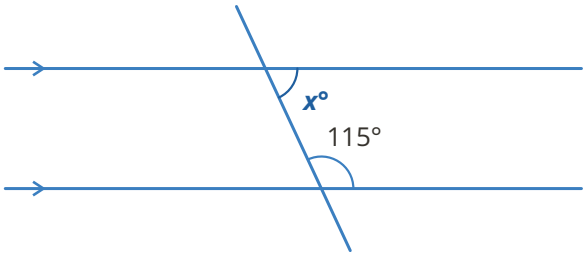
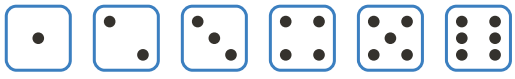
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Convert 3 L to cubic centimetres.	FORMULA $1\ cm^3 = 1\ mL,\ 1\ L = 1000\ mL$ COPY SUBSTITUTE ANSWER
3 Calculate. 60% of 35 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 A kayak tour on Cudgen Creek leaves at 09:15 and returns at 11:50. How long is the tour?	FORMULA Time taken = finish time − start time (60 minutes = 1 hour) COPY SUBSTITUTE ANSWER
5 A fair die is rolled. Find the probability of rolling a 5. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

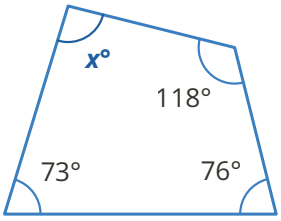
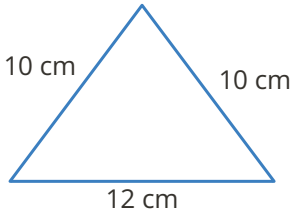
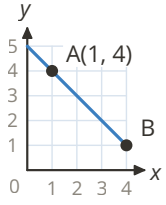
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the perimeter of this triangle. 	FORMULA Perimeter = add the lengths of all the sides COPY SUBSTITUTE ANSWER
3 Calculate. $56.2 \div 10$	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Expand. $2(x + 7)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = 5 - x$. Write the coordinates of B. 	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

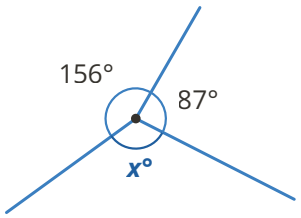
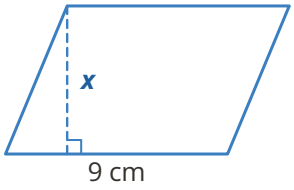
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 3 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 The area of this parallelogram is 54 cm². Find x. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $-12 \div (-4) + 2 \times (-5)$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Simplify. $8 : 20$	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 Find the probability that the marble is not blue. $P(\text{blue}) = \frac{5}{12}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

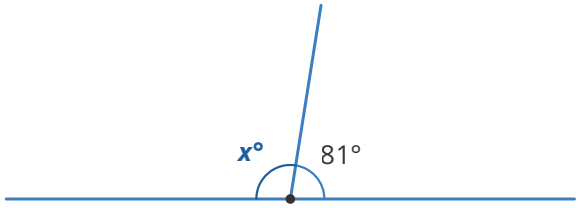
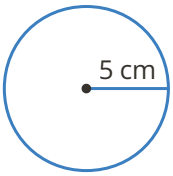
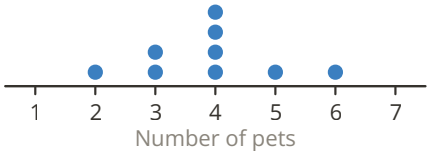
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{9}{10}$ of 70 kg	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Solve. $x + 7 = 15$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 The dot plot shows the number of pets students in a class own. Find the mode. 	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

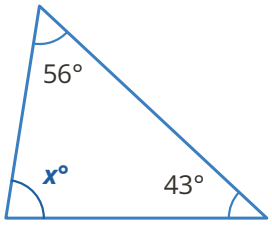
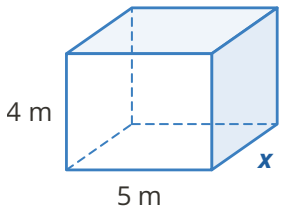
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 The volume of this rectangular prism is 100 m^3. Find x. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 Write as a decimal. 35%	FORMULA Percentage to decimal: percentage ÷ 100 COPY SUBSTITUTE ANSWER
4 Share \$90 in the ratio 2 : 7. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Find the mean of these goal tallies. Goals: 7, 9, 4, 10, 5	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

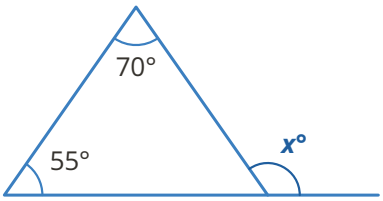
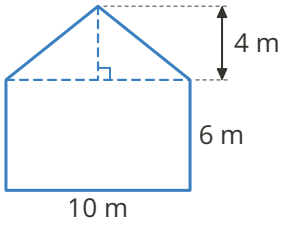
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 4 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 A netball shooter scored 21 goals from 25 shots. What percentage of the shots were goals?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Find the value of this expression when a = 13 and b = 7. $\frac{a+b}{2}$	FORMULA $\frac{a+b}{2}$ COPY SUBSTITUTE ANSWER
5 Find the range of these netball scores. Goals: 26, 18, 31, 22, 24	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

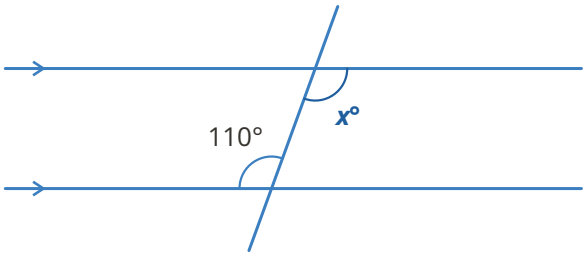
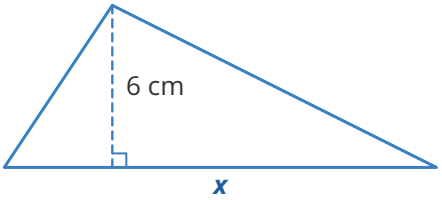
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Alternate angles are equal COPY SUBSTITUTE ANSWER
2 The area of this triangle is 48 cm². Find x.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 40% of $75	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Which bag of potatoes is the better buy? Bag A: 3 kg for $12 Bag B: 5 kg for $15	FORMULA Price per unit = price ÷ number of units; best buy = lowest price per unit COPY SUBSTITUTE ANSWER
5 A spinner was spun 50 times. Find the experimental probability of red. Colour: Red, Blue, Green Frequency: 14, 26, 10	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

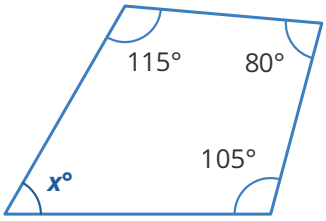
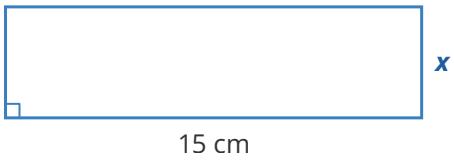
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 The area of this rectangle is 60 cm^2. Find x.  	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER
3 Evaluate. $10^2 + 3^3$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 Expand. $7(2 + k)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is red. Colour: Red, Yellow, Blue Marbles: 2, 3, 5	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

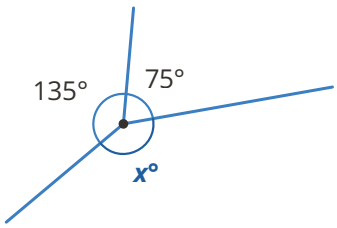
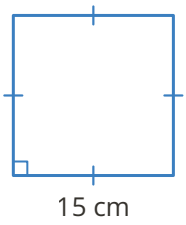
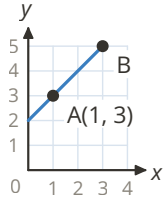
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 5 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the perimeter of this square.  	FORMULA $P = 4s$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{3}{7}$ of 56	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Solve. $7 + 3p = 25$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = x + 2$. Write the coordinates of B.  	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

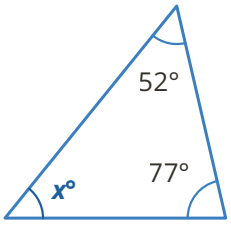
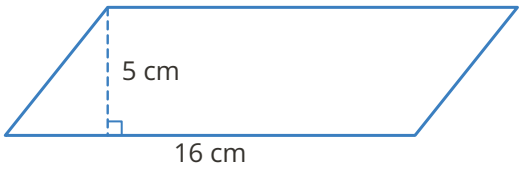
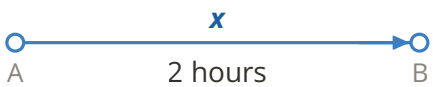
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 15%	FORMULA Percentage to fraction: write over 100, then simplify COPY SUBSTITUTE ANSWER
4 A cyclist rides from A to B at an average speed of 15 km/h. Find x. 	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the probability that the target is not hit. $P(\text{hit}) = \frac{7}{9}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

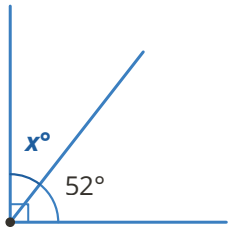
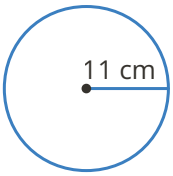
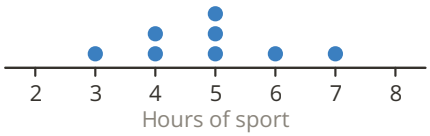
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Complementary angles add to 90° COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER
3 Evaluate. $(-2) \times (-5) - 14$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $y = 9$. $5(y - 2)$	FORMULA $5(y - 2)$ COPY SUBSTITUTE ANSWER
5 The dot plot shows the hours of sport some students played last week. Find the median. 	FORMULA Median = middle score when the scores are in order COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

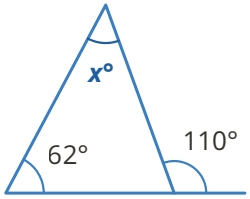
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 6 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Convert 3 m^2 to square centimetres.	FORMULA 1 m^2 = 10 000 cm^2 COPY SUBSTITUTE ANSWER
3 Write as a percentage. 33 out of 60	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Write 11:05 pm in 24-hour time.	FORMULA 24-hour time: pm hours add 12 (1:00 pm = 13:00); 12:00 am = 00:00 COPY SUBSTITUTE ANSWER
5 Find the range of these scores. Scores: 9, 3, 12, 7, 5	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

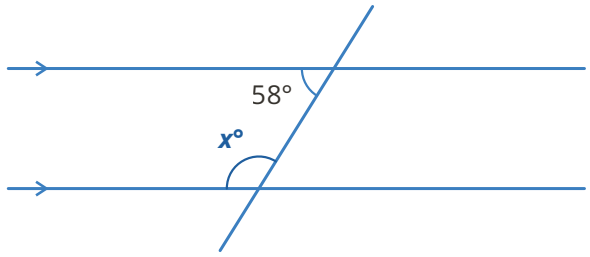
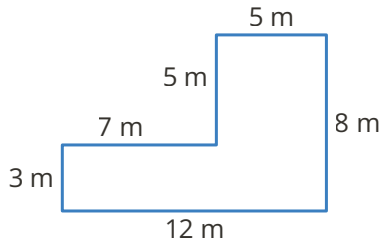
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Co-interior angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the area of this shape.  	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. 4.8×10	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Write as a ratio in simplest form. 45 min : 1 h	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 A netball goal shooter scored 32 goals from 40 shots. Find the experimental probability of scoring a goal.	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

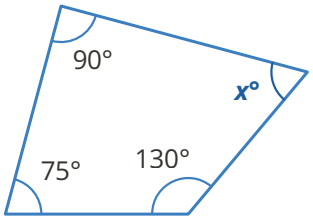
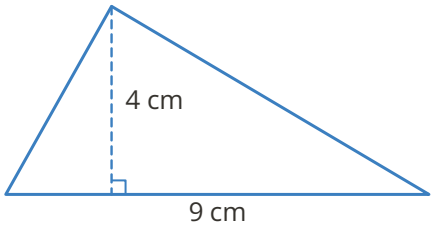
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this triangle.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. 20% of 45 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Expand. $3(2x + 4)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mean of these marks. Marks: 14, 11, 17, 12, 16	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

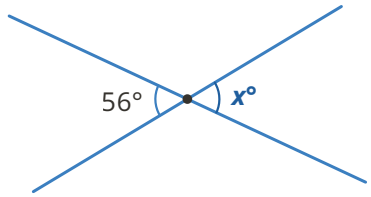
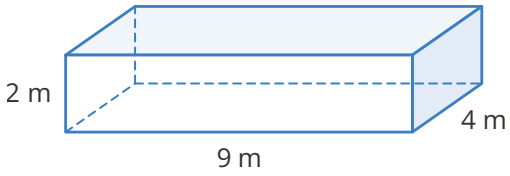
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 7 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Vertically opposite angles are equal COPY SUBSTITUTE ANSWER
2 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER
3 At the Kingscliff markets, $\frac{3}{5}$ of the 40 stalls sell food. How many stalls sell food?	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Share \$120 in the ratio 3 : 5. Give both shares.	FORMULA One part = amount ÷ total number of parts COPY SUBSTITUTE ANSWER
5 Complete the table for $y = 4x$. x: 1, 2, 3, 4 y: 4, , , 16	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

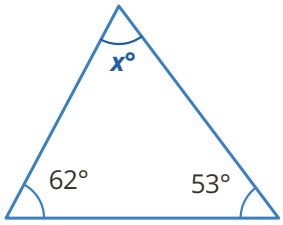
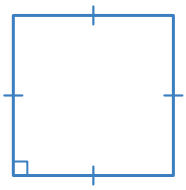
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this square. 	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Write as a percentage. $\frac{2}{5}$	FORMULA Fraction to percentage: fraction × 100% COPY SUBSTITUTE ANSWER
4 Solve. $y - 15 = 8$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A letter is chosen at random from this word. Find $P(S)$. SUCCESS	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

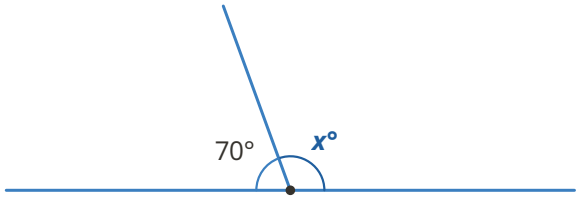
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Find the perimeter of this rectangle.  	FORMULA $P = 2(l + w)$ COPY SUBSTITUTE ANSWER
3 Evaluate. $\sqrt{49} + 2^3$	FORMULA $a^n = a \times a \times \ldots \times a$ (n times); $\sqrt{a^2} = a$, $\sqrt[3]{a^3} = a$ COPY SUBSTITUTE ANSWER
4 A tap fills a tank with 240 L of water in 8 minutes. Find the rate in litres per minute.	FORMULA Unit rate = amount ÷ number of units COPY SUBSTITUTE ANSWER
5 Find the probability that it does not rain. $P(\text{rain}) = \frac{2}{5}$	FORMULA $P(\text{not E}) = 1 - P(\text{E})$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

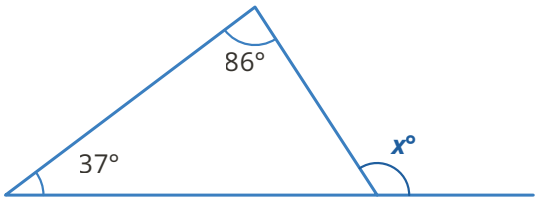
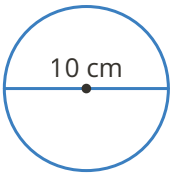
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 8 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the circumference of this circle, correct to 1 decimal place.  	FORMULA $C = \pi d$ COPY SUBSTITUTE ANSWER
3 24 of the 30 students in a class walk or ride to school. What percentage is that?	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Find the value of this expression when n = 4. $\frac{12}{n} + 3$	FORMULA $\frac{12}{n} + 3$ COPY SUBSTITUTE ANSWER
5 Find the range of these scores. Scores: 40, 38, 45, 33, 41	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

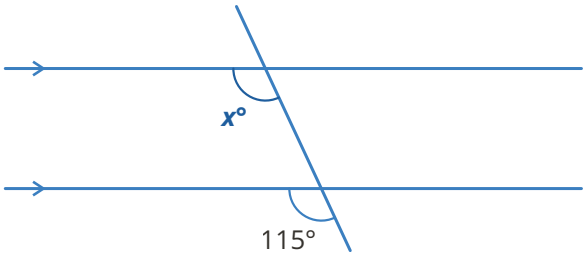
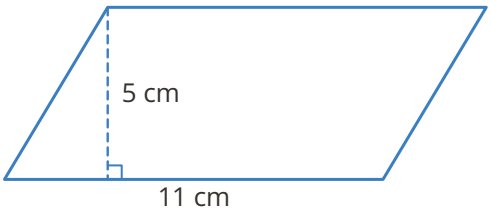
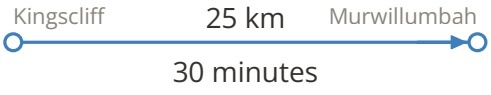
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Corresponding angles are equal COPY SUBSTITUTE ANSWER
2 Find the area of this parallelogram.  	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER
3 Evaluate. $-5 \times 4 + 6 \times 3$	FORMULA Brackets first, then × and ÷ from left to right, then + and − from left to right COPY SUBSTITUTE ANSWER
4 A car travels from Kingscliff to Murwillumbah. Find the average speed in km/h.  	FORMULA Speed = $\frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER
5 Find the mode of these scores. 2 4 4 1 3 2 4 5	FORMULA Mode = the most common score COPY SUBSTITUTE ANSWER

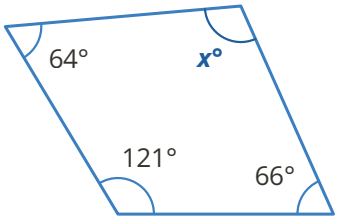
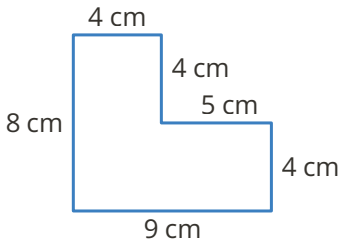
KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

Week 9 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360°$ COPY SUBSTITUTE ANSWER
2 Find the area of this shape. 	FORMULA Area of a composite shape = add (or subtract) the areas of its parts COPY SUBSTITUTE ANSWER
3 Calculate. 90% of 50 kg	FORMULA Percentage of a quantity = percentage ÷ 100 × quantity COPY SUBSTITUTE ANSWER
4 Expand. $3(4p + 1)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER
5 Find the mean of these temperatures. °C: 24, 31, 27, 30, 28	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

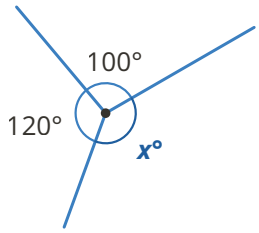
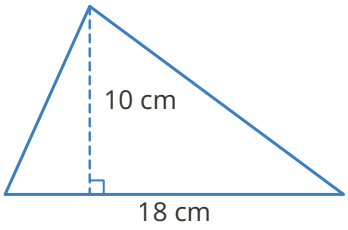
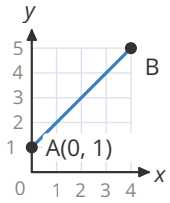
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 9 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER
2 Find the area of this triangle.  	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER
3 Calculate. $\frac{3}{4}$ of 60	FORMULA Fraction of a quantity = fraction × quantity COPY SUBSTITUTE ANSWER
4 Solve. $3x + 5 = 20$	FORMULA Do the same to both sides COPY SUBSTITUTE ANSWER
5 A and B lie on the line $y = x + 1$. Write the coordinates of B.  	FORMULA Substitute each x into the rule COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

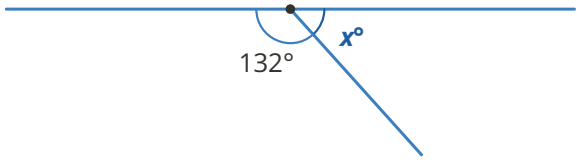
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 1

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Supplementary angles add to 180° COPY SUBSTITUTE ANSWER
2 Convert 7 km to metres.	FORMULA 1 cm = 10 mm, 1 m = 100 cm, 1 km = 1000 m COPY SUBSTITUTE ANSWER
3 Write as a fraction in simplest form. 0.48	FORMULA Decimal to fraction: write over 10, 100 or 1000, then simplify COPY SUBSTITUTE ANSWER
4 A movie starts at 19:35 and finishes at 21:20. How long is the movie?	FORMULA Time taken = finish time − start time (60 minutes = 1 hour) COPY SUBSTITUTE ANSWER
5 A die was rolled 60 times. Find the experimental probability of rolling a 6. Number: 1, 2, 3, 4, 5, 6 Frequency: 8, 11, 9, 12, 10, 10	FORMULA Experimental probability = frequency ÷ number of trials COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

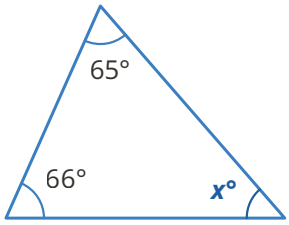
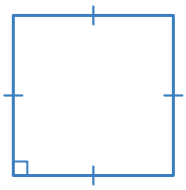
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 2

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA $a + b + c = 180°$ COPY SUBSTITUTE ANSWER
2 Find the area of this square.  	FORMULA $A = s^2$ COPY SUBSTITUTE ANSWER
3 Calculate. 260 ÷ 1000	FORMULA × 10, 100, 1000: digits move 1, 2, 3 places left; ÷ 10, 100, 1000: they move right COPY SUBSTITUTE ANSWER
4 Simplify. 10 : 25 : 15	FORMULA Divide both parts by their highest common factor COPY SUBSTITUTE ANSWER
5 One marble is chosen at random from this bag. Find the probability that it is white. Colour: Red 5, White 3	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER

KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

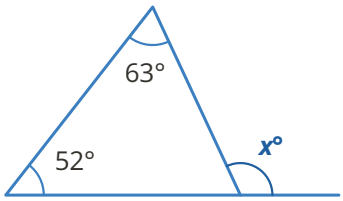
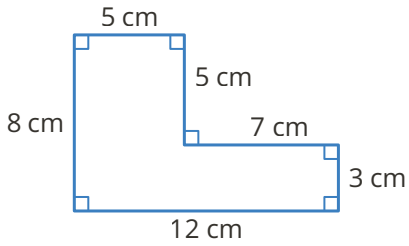
MATHEMATICS · YEAR 8 · THE FRACTAL FIVE

Week 10 · Session 3

TERM 4 Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x.  	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER
2 Find the perimeter of this shape.  	FORMULA Perimeter = add the lengths of all the sides COPY SUBSTITUTE ANSWER
3 Write as a percentage. 16 out of 64	FORMULA Percentage = part ÷ whole × 100% COPY SUBSTITUTE ANSWER
4 Find the value of this expression when $c = 3$ and $d = -5$. $7c - d$	FORMULA $7c - d$ COPY SUBSTITUTE ANSWER
5 Find the range of these heights. cm: 104, 96, 118, 101, 99	FORMULA Range = highest − lowest COPY SUBSTITUTE ANSWER